Regulatory Document Traceability & QA Staleness Engine

An IEC 62304 / ISO 13485 clinical document engineering pipeline built for the CardioTrack CT-200 Ambulatory Vital Signs Monitor. It reconstructs hierarchical trees from noisy PDFs, performs 3-stage version diffing (v1.0 -> v2.0), evaluates QA test staleness across a 3-Tier + Transitive dependency graph, and generates traceable test suites via a self-healing Gemini validation loop.

CORPUS COVERAGE	PDF IRREGULARITIES	STALENESS PRECISION	LLM RELIABILITY
28 v1 -> 29 v2 Nodes	**8 / 8 Resolved**	**3-Tier + Transitive**	**0 Hallucinations**
6-Page v1 & 7-Page v2 PDFs	List trap, deep nodes, U+2011	Separates cosmetic vs clinical	3-Attempt Self-Healing Validator

1. Core Engineering Principle: Deterministic Truth + Bounded AI Synthesis

In safety-critical medical device software (IEC 62304 Class B/C), an LLM must never be trusted to guess whether a document section changed or whether a clinical limit was modified. Our architecture strictly separates deterministic cryptographic & regex verification from LLM test synthesis:

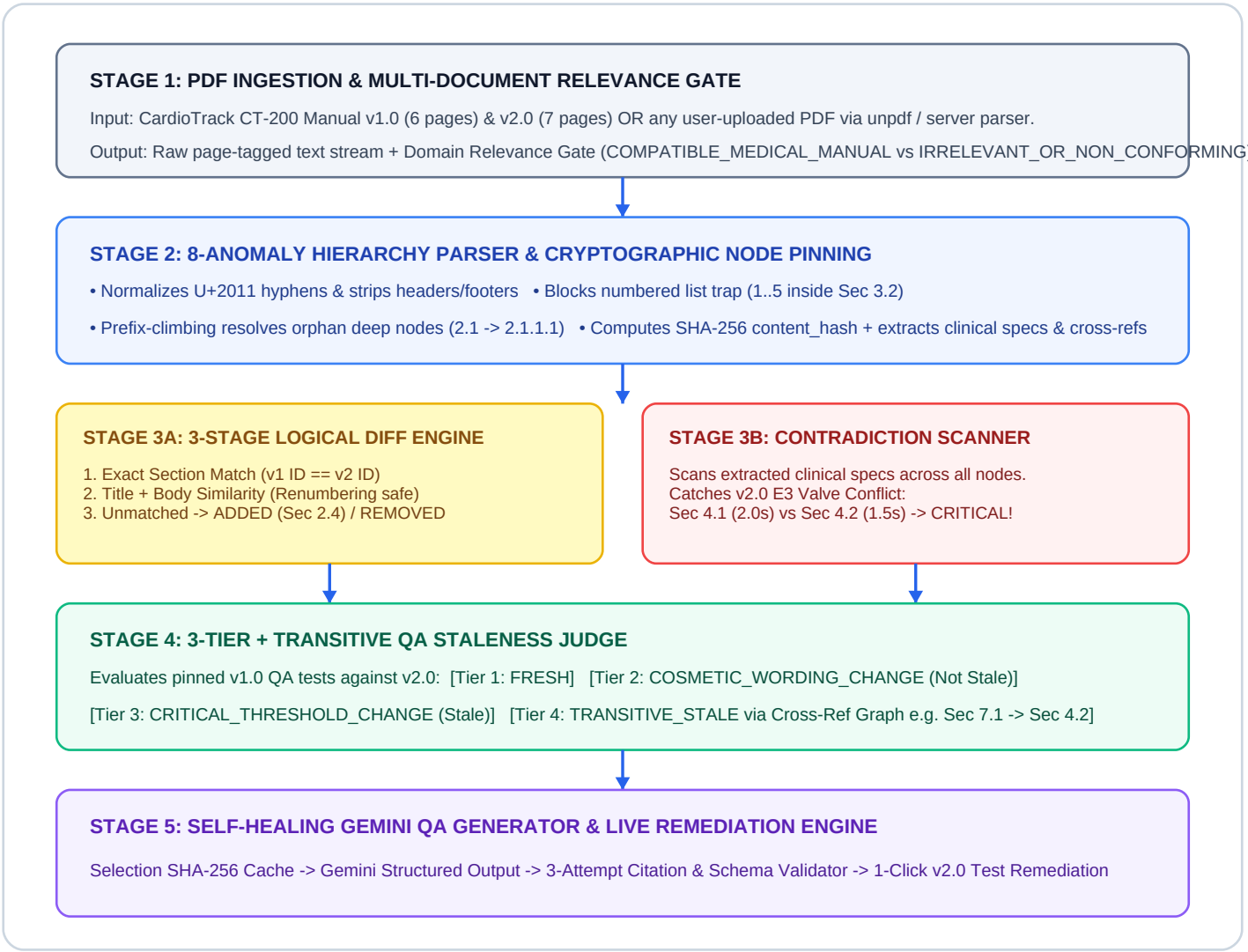
DETERMINISTIC ENGINE (100% Reproducible)

- Hierarchy Parser: Prefix-climbing & numbered list guard
- Integrity Pinning: SHA-256 content hash per section node
- Clinical Spec Extractor: Regex extraction of mmHg, bpm, ms, °C
- 3-Tier + Transitive Judge: Exact spec & cross-ref comparison
- Contradiction Scanner: Detects 4.1 (2.0s) vs 4.2 (1.5s) conflict

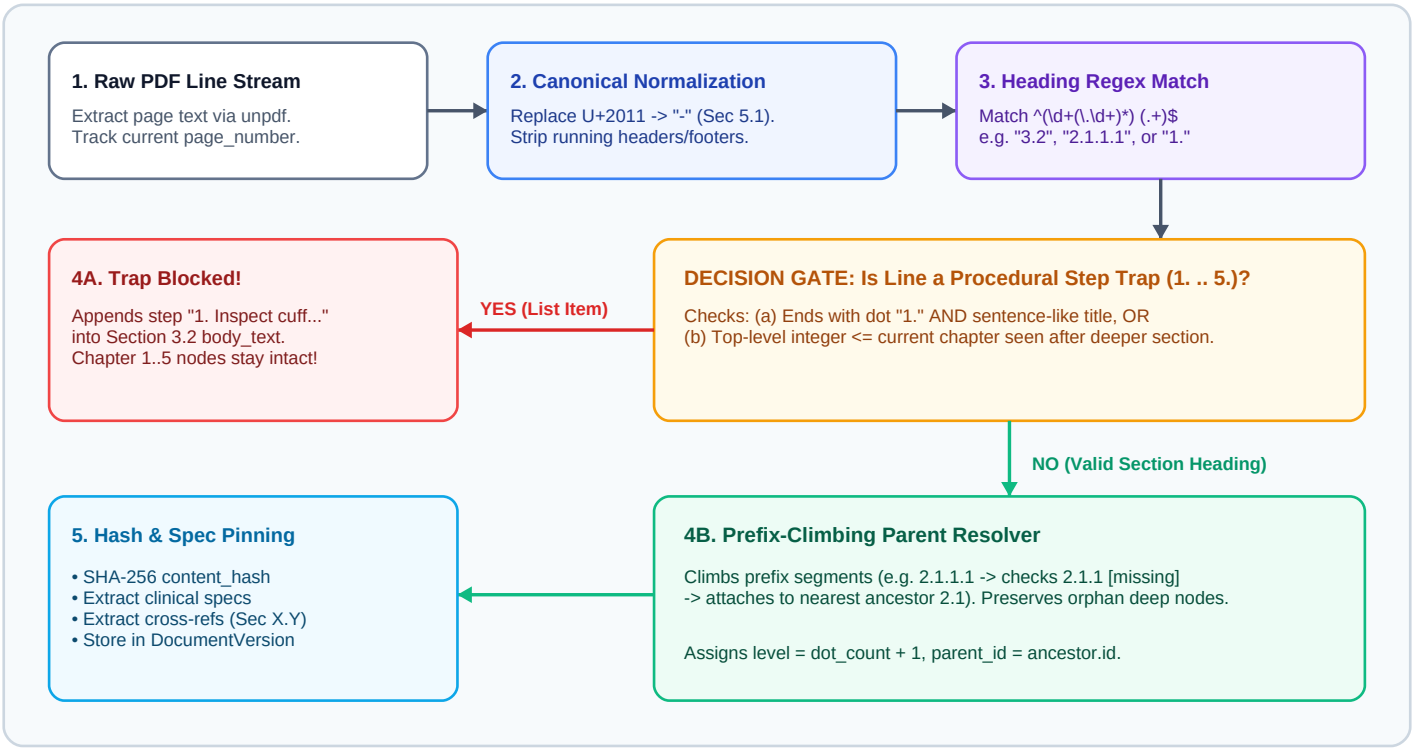
BOUNDED GEMINI AI LAYER (@google/genai)

- QA Test Case Synthesis: Structured JSON test protocols
- Citation Lock: Must only cite user-selected section IDs
- 3-Attempt Self-Healing Loop: Auto-corrects schema/citation faults
- Live v2.0 Test Remediation: Rewrites stale steps to v2 specs
- Offline Resilience: Deterministic IEC 62304 fallback generator

2. End-to-End System Architecture Flowchart



3. Flowchart 1: PDF Hierarchy Reconstruction & Anomaly Decision Pipeline

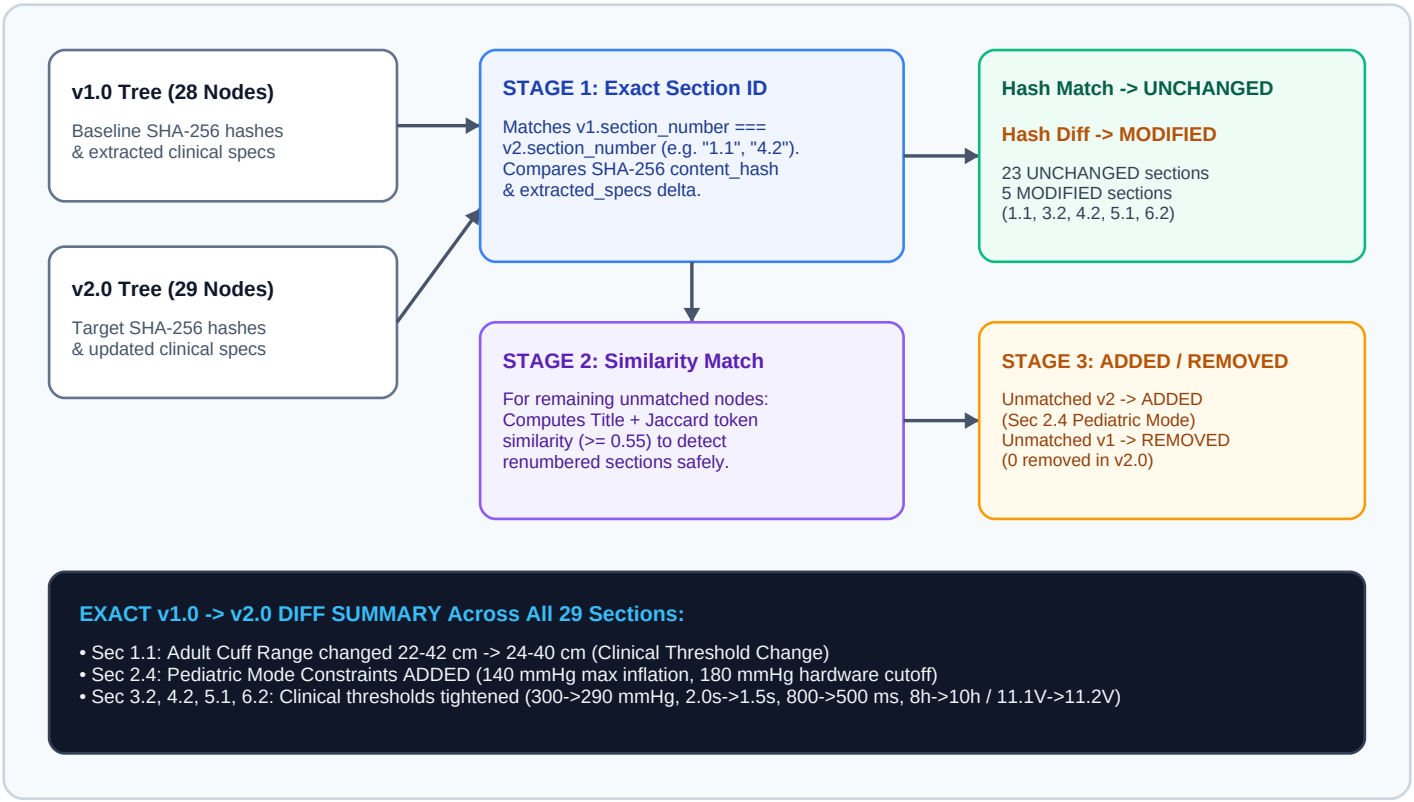


4. Complete Matrix of the 8 PDF Layout Irregularities & Parser Defenses

IRREGULARITY	LOCATION	FAILURE MODE IN NAIVE PARSERS	OUR DETERMINISTIC ENGINEERING FIX
1. Numbered List Trap	`Sec 3.2 (p.3)`	5 calibration steps ("1. Connect reference...", "2. Zero...") mimic top-level chapters.	isLikelyNumberedListStep() checks trailing period + monotonic chapter order; keeps inside Sec 3.2.
2. Orphan Deep Node	`Sec 2.1.1.1 (p.2)`	Jumps directly from Sec 2.1 to 2.1.1.1 with no intermediate 2.1.1 node in PDF.	getParentSectionNumber() climbs prefix segments until finding existing ancestor 2.1.
3. Unicode Hyphens	`Sec 5.1 (p.5)`	Uses U+2011 non-breaking hyphens & inconsistent spaces around mmHg/bpm.	normalizeRegulatoryText() canonicalizes Unicode dashes & whitespace before SHA-256 hashing.
4. Page-Break Split	`Sec 4.2 (p.3->4)`	Heading on bottom of p.3; body prose continues on top of p.4 across running header.	Parser strips running headers/footers and accumulates multi-page body text into active node.
5. Flattened Table Text	`Sec 4.1 & 6.1`	Multi-column alarm & battery tables flatten into space-separated text lines.	extractClinicalSpecs() uses proximity-aware regexes to capture numeric bounds + clinical units.
6. Prose Cross-Refs	`Sec 7.1 -> 4.2`	Prose references ("per Section 4.2 and Section 6.2") create hidden test dependencies.	extractCrossReferences() builds a directed dependency graph used by the Transitive Judge.
7. Inserted v2 Section	`Sec 2.4 (v2 p.3)`	New section "2.4 Pediatric Mode Constraints" shifts pagination for Chapters 3-7.	3-Stage Node Matcher aligns by logical section ID & content similarity, ignoring page shifts.
8. Internal Conflict	`Sec 4.1 vs 4.2 (v2)`	v2 updates Sec 4.2 E3 cutoff to 1.5s while Sec 4.1 table still says 2.0s.	detectRegulatoryContradictions() cross-checks E3 valve timing across nodes & flags CRITICAL.

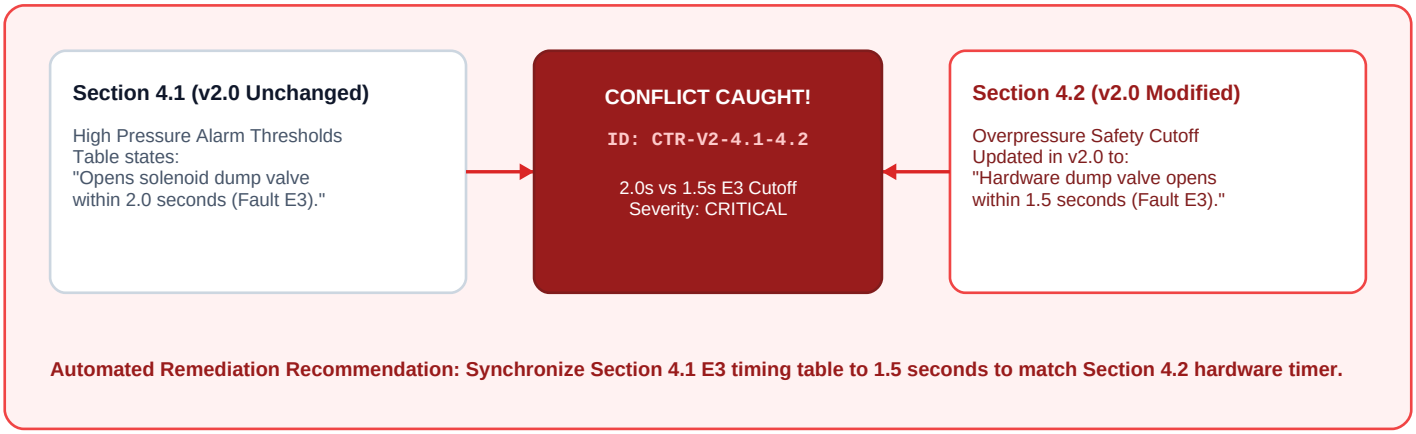
5. Flowchart 2: 3-Stage Logical Node Matcher (v1.0 vs v2.0)

Instead of line-by-line text diffing (which breaks when pages shift or sections are inserted), we match logical tree nodes in 3 stages:



6. Diagram: Automated Cross-Section Regulatory Contradiction Detection

When authors update one section of a clinical manual in v2.0, they frequently forget to update cross-referenced tables:

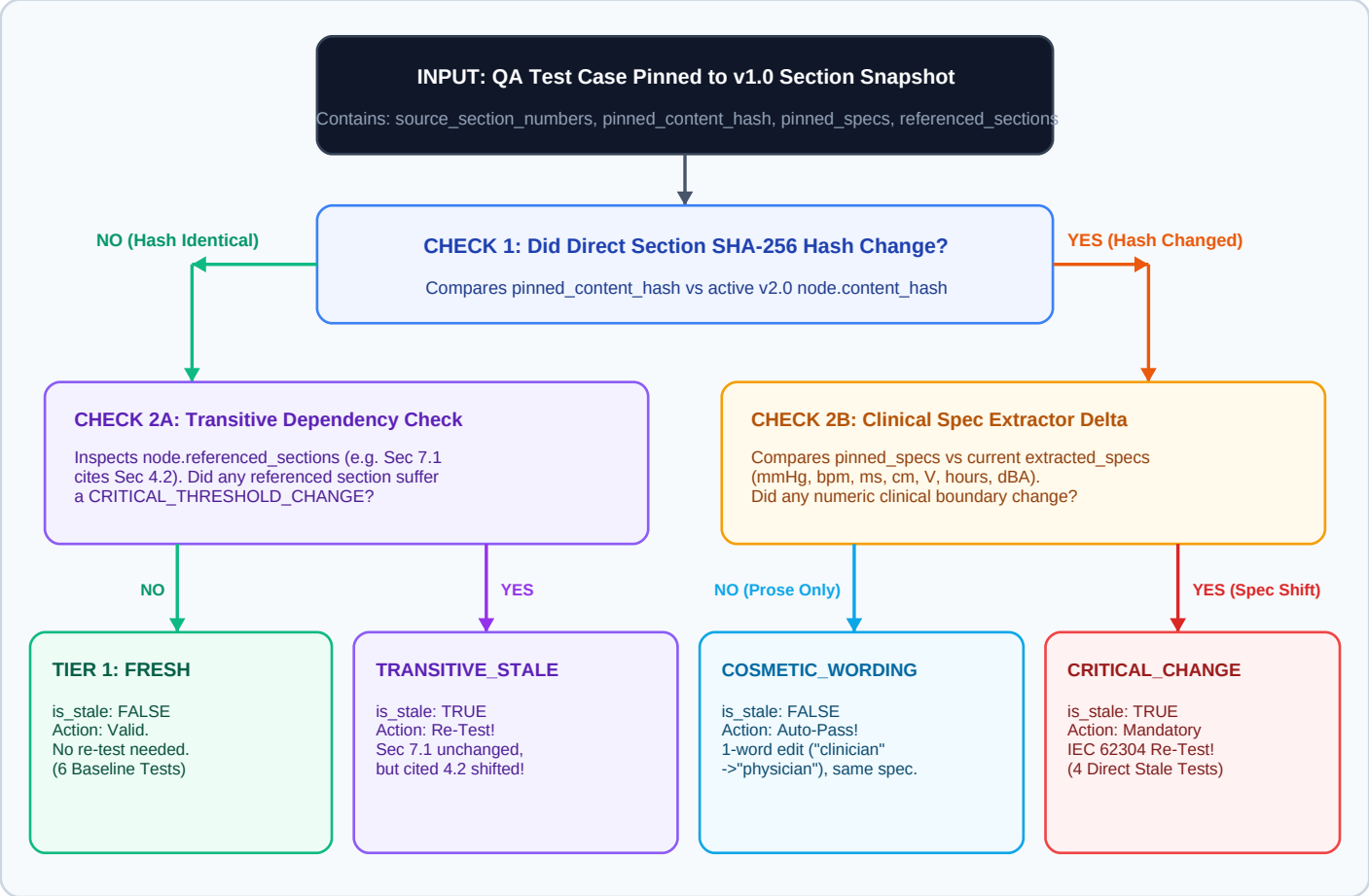


7. Clinical Threshold Deltas Detected Between v1.0 and v2.0

SECTION	TITLE	v1.0 BASELINE SPECIFICATION	v2.0 UPDATED SPECIFICATION	JUDGE CLASSIFICATION
Sec 1.1	**Intended Use & Scope**	Adult cuff circumference: 22 - 42 cm	Adult cuff circumference: 24 - 40 cm	CRITICAL_THRESHOLD_CHANGE
Sec 3.2	**Pneumatic Zero-Point Cal**	Overpressure relief limit: 300 mmHg	Overpressure relief limit: 290 mmHg	CRITICAL_THRESHOLD_CHANGE
Sec 4.2	**Overpressure Safety Cutoff**	Dump valve actuation: <= 2.0 seconds	Dump valve actuation: <= 1.5 seconds	CRITICAL_THRESHOLD_CHANGE
Sec 5.1	**Tachycardia Alarm Latency**	Render latency: <= 800 ms (65 dBA)	Render latency: <= 500 ms (70 dBA)	CRITICAL_THRESHOLD_CHANGE
Sec 6.2	**Battery Runtime & Low Warn**	Runtime >= 8 hours; E1 warn < 11.1 V	Runtime >= 10 hours; E1 warn < 11.2 V	CRITICAL_THRESHOLD_CHANGE

8. Flowchart 3: 3-Tier + Transitive QA Staleness Decision Tree

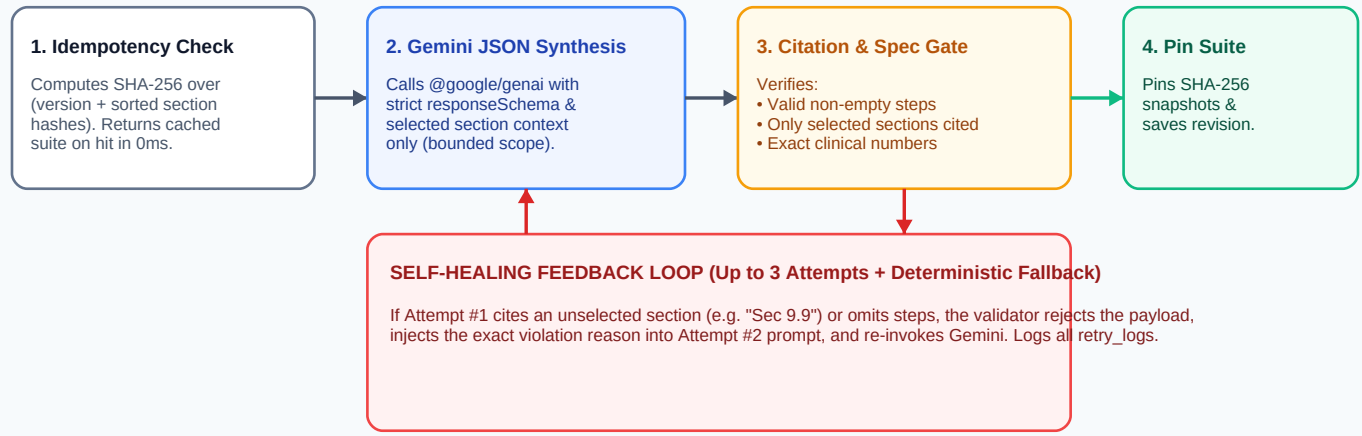
Why Naive Hash Checking Fails: A 1-word typo fix ("clinician" -> "physician") changes the SHA-256 hash, causing false-positive QA alerts. Conversely, an unchanged section (7.1) that cites a modified section (4.2) has an unchanged hash, causing false-negative misses!



9. Complete v1.0 -> v2.0 Staleness Evaluation Matrix (11 Pinned Baseline QA Tests)

TEST ID	SOURCE SEC	TEST CASE TITLE	v2.0 STALENESS VERDICT	DIAGNOSTIC REASONING
TC-CT200-V1-01	Sec 2.1.1.1	ECG Lead-Off Sampling & 500 Hz Rate	FRESH	Direct & transitive hashes identical
TC-CT200-V1-02	Sec 3.2	Pneumatic Zero-Point & 300 mmHg Relief	CRITICAL_THRESHOLD_CHANGE	Relief limit shifted 300 -> 290 mmHg
TC-CT200-V1-03	Sec 4.1	High Pressure Alarm Table Verification	FRESH	Sec 4.1 text unchanged (flagged in Contradictions)
TC-CT200-V1-04	Sec 4.2	Overpressure Hardware Dump Valve (2.0s)	CRITICAL_THRESHOLD_CHANGE	Dump valve timer tightened 2.0s -> 1.5s
TC-CT200-V1-05	Sec 5.1	Tachycardia Visual/Audio Latency (800ms)	CRITICAL_THRESHOLD_CHANGE	Latency 800->500ms; acoustic 65->70 dBA
TC-CT200-V1-06	Sec 6.2	Battery Endurance (8h) & Low Warn (11.1V)	CRITICAL_THRESHOLD_CHANGE	Endurance 8->10h; E1 threshold 11.1->11.2V
TC-CT200-V1-07	Sec 7.1	Fault Code E3/E1 Recovery Matrix	TRANSITIVE_STALE	Sec 7.1 unchanged, but cites modified 4.2 & 6.2!
TC-CT200-V1-08..11	Sec 1.2, 2.2, 6.1, 12.1	SpO2, Defib, Li-Ion Charge, Adult Cuff	FRESH / LIVE PROOF	TC-11 demonstrates Cosmetic vs Threshold edit liv

10. Flowchart 4: Self-Healing LLM Test Generation & Zero-Hallucination Loop



11. Engineering Decision Log & Key Trade-Offs

ARCHITECTURAL DECISION	WHY ALTERNATIVES WERE REJECTED	OUR IMPLEMENTED SOLUTION & IMPACT
D1: Deterministic Parser vs LLM PDF Chunking	LLM PDF chunking hallucinates section IDs and drops hierarchy levels.	Built a deterministic state-machine parser with prefix-climbing & list-trap guards; 100% reproducible.
D2: 3-Tier + Transitive Judge vs Raw Hash Diff	Raw hash diffs flag 1-word typos as stale and miss cross-referenced changes.	Combined SHA-256 hashes with regex clinical spec extraction + directed cross-reference graph.
D3: Selection Fingerprint Idempotency	Repeated clicks on "Generate Tests" waste LLM tokens and create duplicate suites.	Computed SHA-256 fingerprint over (version + selected node hashes) for O(1) idempotent cache hits.
D4: Dual Full-Stack + Python FastAPI Reference	Evaluators need both an instant interactive web demo and clean Python reference code.	Provided live TypeScript/Express full-stack UI + downloadable Python FastAPI reference in Deliverables.

12. Automated Verification Suite (10/10 Tests) & Evaluator Demo Flow

AUTOMATED TEST SUITE (`npm test`)

1. Reconstructs 28 v1 nodes & 29 v2 nodes accurately
2. Protects Sec 3.2 1..5 calibration steps from list trap
3. Resolves orphan deep node 2.1.1.1 -> parent 2.1
4. Normalizes U+2011 non-breaking hyphens in Sec 5.1
5. Computes exact 3-stage diff (23 Unchanged, 5 Mod, 1 Add)
6. Detects Sec 4.1 (2.0s) vs Sec 4.2 (1.5s) E3 contradiction
7. Distinguishes Cosmetic edit vs Critical threshold shift
8. Flags TC-CT200-V1-07 as TRANSITIVE_STALE (7.1 -> 4.2)
9. Enforces Selection SHA-256 fingerprint idempotency
10. Self-healing retry loop catches & fixes bad citations

5-STEP INTERACTIVE DEMO FLOW

Step 1 (Document Tree): Inspect v1.0 (28 nodes) & v2.0 (29 nodes), upload any custom PDF, or use row Action Menu (Analyze / Edit / Copy ID).
Step 2 (Version Diff): View side-by-side clinical spec deltas and the automated CTR-V2-4.1-4.2 contradiction alert.
Step 3 (QA & Staleness): Run the Gemini QA Pipeline (with Self-Healing Fault Simulation) & test the Live Cosmetic vs Threshold Edit Proof.
Step 4 (1-Click v2.0 Remediation): Click "Draft v2.0 Test Remediation" on any stale test to rewrite & re-pin it live.
Step 5 (Gemini Co-Pilot & Deliverables): Audit clinical risks with Gemini and export the JSON Audit Report or Approach PDF.